

Welcome to the BSPSSEPy Program

Version: 0.6

Last Updated: 14 Aug 2025

Developed by: Ilyas Farhat

Contact: ilyas.farhat@outlook.com

Copyright (c) 2024–2025, Ilyas Farhat

All rights reserved.

Overview

BSPSSEPy Application is a Black-Start simulation tool based on PSSE Power Simulator, that uses the Python PSSE Library to run these simulations. BSPSSEPy extends the dynamic simulation capabilities of PSSE to model:

- Dynamic Black-Start behavior (phases) of non-black-start generators. It models their cranking phase, ramp-up phase (if the generator governor model does not model ramping limits).
 - Can execute a black-start plan and apply AGC control during the restoration process.
 - Dynamically tracks the elements in the network and monitor voltages and frequencies.
-

Goals in Version 0.6 [in progress]

- Generating new documentation following the new structure of the program **[priority]**
 - `manual.md` is the user guide manual on how to run simulations using BSPSSEPy application. To view the document, use MD viewer like: **Obsidian**, **VS Code** or browser extensions such as **Markdown Viewer** (**recommended**) or other softwares that can render md files.
 - `manual.pdf` and `manual_A3.pdf` are pdf exports from `manual.md`, but *links are broken in these files.*
 -
 - Continue migrating all measurements updates to the new function *[paused]*.
 - Continue rewriting application code following PEP8 standard *[paused]*.
-

Changes in Version 0.5

- Rewriting the whole application code following PEP 8 standard [partially done].
 - Grouping Measurements in one main function group for all elements ~~in progress~~ incomplete and paused to work on documentation].
 - ~~Generating new documentation following the new structure~~ [not started -> moved to Version 0.6].
 - Extending the functionality to include BESS modeling and control [Completed !].
-

- Fixing minor GUI glitches and upgrading the dependencies if needed [partially done].
-

Changes in Version 0.4

- Added BSPSSEPy Dashboard based on the `textual` Python library.
 - Made many changes to how the simulation works and how the plan is being read.
 - Introduced new flags for greater control over action execution.
 - Added new function `GenUpdate` to control output of non-AGC generators.
 - Fixed various issues in the code.
-

Changes in Version 0.3

- Fixed `BSPSSEPySim` class to address generator measurement channel issues.
 - Updated documentation for `BSPSSEPySim`.
-

Changes in Version 0.2.1

- Successfully modeled the cranking phase of generators.
 - Refactored `BSPSSEPyGenFunctions` and extended the generator dataframe with phase info.
-

Changes in Version 0.2

- Rewrote all classes to align with the updated dataframe structure (`BSPSSEPyGen`, `BSPSSEPyBrn`, `BSPSSEPyTrn`, `BSPSSEPyLoad`, `BSPSSEPyBus`).
- Updated some associated markdown documentation (to be revisited in v0.3).